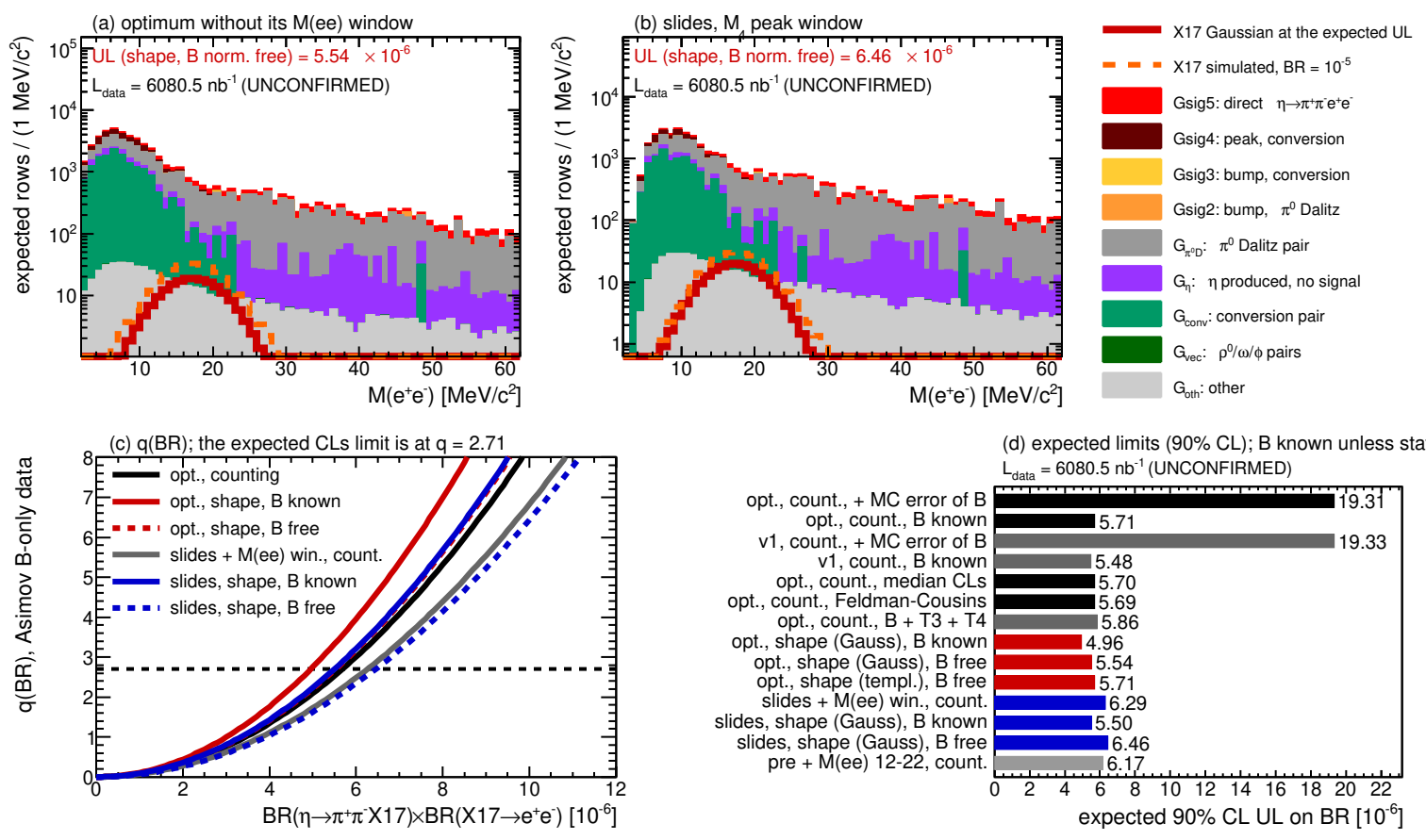
(a) optimum without its M(ee) window
UL (shape, B norm. free) = 5.54 × 10^-6
L_data = 6080.5 nb^-1 (UNCONFIRMED)
expected rows / (1 MeV/c^2)
M(e+e-) [MeV/c^2]
(b) slides, M_4 peak window
UL (shape, B norm. free) = 6.46 × 10^-6
L_data = 6080.5 nb^-1 (UNCONFIRMED)
expected rows / (1 MeV/c^2)
M(e+e-) [MeV/c^2]
X17 Gaussian at the expected UL
X17 simulated, BR = 10^-5
Gsig5: direct η→π+π-e+e-
Gsig4: peak, conversion
Gsig3: bump, conversion
Gsig2: bump, π0 Dalitz
G_π0D: π0 Dalitz pair
G_η: η produced, no signal
G_conv: conversion pair
G_vec: ρ0/ω/ϕ pairs
G_oth: other
(c) q(BR); the expected CLs limit is at q = 2.71
q(BR), Asimov B-only data
opt., counting
opt., shape, B known
opt., shape, B free
slides + M(ee) win., count.
slides, shape, B known
slides, shape, B free
BR(η→π+π-X17)×BR(X17→e+e-) [10^-6]
(d) expected limits (90% CL); B known unless sta
L_data = 6080.5 nb^-1 (UNCONFIRMED)
opt., count., + MC error of B 19.31
opt., count., B known 5.71
v1, count., + MC error of B 19.33
v1, count., B known 5.48
opt., count., median CLs 5.70
opt., count., Feldman-Cousins 5.69
opt., count., B + T3 + T4 5.86
opt., shape (Gauss), B known 4.96
opt., shape (Gauss), B free 5.54
opt., shape (templ.), B free 5.71
slides + M(ee) win., count. 6.29
slides, shape (Gauss), B known 5.50
slides, shape (Gauss), B free 6.46
pre + M(ee) 12-22, count. 6.17
expected 90% CL UL on BR [10^-6]